

CLARA DIMARCO

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RESEARCH INTERESTS

My research operates at the intersection of AI in Education (AIED), Learning Sciences, and Human-Computer Interaction (HCI). My work builds and evaluates adaptive, human-centered systems, with a focus on the reliability of LLM-generated instruction. Driven by a foundational background in mathematics and logic, my core research investigates intelligent tutoring systems for propositional logic and the formal evaluation of LLM-generated pedagogical proofs and hints. Complementing this technical core, I design interactive educational interfaces and lead K–12 AI-literacy and accessible-computing outreach for students with disabilities. Across domains spanning educational technology, aerospace, and civil infrastructure, my work is grounded in a consistent standard: whether a system is genuinely accessible, usable, and effective for the people it's built for.

PUBLICATIONS

Journal Articles

[1] Sutapa Dey Tithi, Arun Kumar Ramesh, **Clara DiMarco**, Xiaoyi Tian, Nazia Alam, Kimia Fazeli, and Tiffany Barnes. 2025. The Promise and Limits of LLMs in Constructing Proofs and Hints for Logic Problems in Intelligent Tutoring Systems. *Computers and Education: Artificial Intelligence* 9 (2025), 100490.

<https://doi.org/10.1016/j.caeai.2025.100490>

Conference Proceedings

[2] Xiaoyi Tian, Yasitha Rajapaksha, Ally Limke, **Clara DiMarco**, Emily Bryans Dobar, Marnie Hill, Jamie Payton, and Tiffany Barnes. 2026. AI Scholars Program: Scaling AI Literacy Through K–12 Outreach. In *Proceedings of the AAAI Conference on Artificial Intelligence (EAAI Track)*, Vol. 40. 40704–40712.

<https://doi.org/10.1609/aaai.v40i47.41518>

EDUCATION

B.S. Computer Science, Minor in Business Administration

Expected Dec. 2026

North Carolina State University | GPA: 3.84

Honors: STARS AI Scholars | Grand Challenges Scholars Program Scholar: Advance Personalized Learning

Study Abroad | Computer Science Coursework | *The Hong Kong Polytechnic University*

Spring 2026

Study Abroad | Sustainable Infrastructure | *Université Côte d'Azur, Nice France*

Spring 2025

B.S. Mathematics, Magna Cum Laude | *University at Albany, SUNY*

May 2022

Relevant Coursework: Intro to AI (in progress), Intro to Logic, Intro to Proofs, Probability, Linear Algebra, Statistical Inference

HCI & LEARNING SCIENCES RESEARCH

Project Lead

Sep. 2026 – Present

“Can You Learn Privately?: Privacy, Access, and Personalized Learning for Tor Users”, NC State University

- Proposed and planned an original research question and methodology from an open-ended course brief, extending a Tor .onion privacy audit into an HCI evaluation of educational access for privacy-reliant learners.
- Lead a 3-person team designing a two-rater heuristic evaluation (Nielsen & Molich) with inter-rater reliability reporting (Cohen's kappa) to identify whether personalized-learning and scholarly platforms make privacy compromises and access blocks perceivable to learners who rely on Tor.

Mobile App Developer & Researcher

Jan. 2025 – Apr. 2025

Game2Learn Lab GCSP REU (PI: Dr. Tiffany Barnes), NC State University

- Conceived and independently developed a prototype of Proof-42, a cross-platform mobile logic tutor (React Native, Tailwind CSS, Supabase), designed from lab survey data showing half of students found the existing tutor's interface their biggest hurdle.
- Reviewed existing logic tutors to design a seven-level scaffolded progression grounded in cognitive-load principles, with screen-reader support and color-blind palettes.
- Conducted iterative peer usability testing to refine navigation and interactive logic elements.

AI & Logic Systems Researcher

May 2024 – Jul. 2024

Game2Learn Lab GCSP REU (PI: Dr. Tiffany Barnes), NC State University

- Co-developed a research question on optimal LLM prompting strategies for logic-proof generation; conducted a systematic literature review and designed standardized prompt templates.
- Introduced bidirectional prompting (FS B-L-DCoT), merging forward and backward reasoning, and compared five few-shot chain-of-thought techniques on GPT-4, grading step-level accuracy in the Deep Thought tutor; implemented a DNF-based proof solver.
- Co-authored the resulting manuscript [1], credited with conceptualization, methodology, validation, and writing (original draft and review & editing).

HUMAN CENTERED SYSTEMS ENGINEERING

Software Engineer, Senior Design Capstone

Aug. 2026 – Present

Rapid Post-Earthquake Bridge Assessment Tool | NC State CCEE Earthquake Research Team & AKDOT

- Engineer a full stack earthquake response tool (Python, FastAPI, SQLAlchemy, MySQL, Docker, Nginx, React, TypeScript, Tailwind CSS) used by AKDOT engineers to interpret and act on bridge integrity data after seismic events.
- Translate complex structural-assessment algorithms from legacy MATLAB models into Python, integrating real-time USGS earthquake data and ground-motion models to automate damage prediction across Alaska's bridge inventory.
- Contribute to the project's ethical design review, evaluating how non-expert field engineers would interpret and trust an automated inspection-priority ranking.

Software Engineer Intern

Jun. 2025 – Aug. 2025

NASA Jet Propulsion Laboratory, Pasadena, CA

- Engineered a JSON-driven configuration web interface that unified fragmented multi-system workflows into a scalable environment for Deep Space Network subsystem engineering teams.
- Automated backend file workflows using Python and Perl CGI scripts, removing manual terminal generation tasks for cross-functional operators.

Systems Engineer

May 2024 – Aug. 2024

NASA L'SPACE Mission Concept Academy | Interdisciplinary Lunar Mission Design

- Co-designed a 199-page lunar water-ice detection mission design with a cross-disciplinary, nationwide student team, defining system requirements, a Science Traceability Matrix, and a Failure Modes and Effects Analysis (FMEA).

TEACHING, OUTREACH & INCLUSIVE DESIGN

Teaching Assistant, Introduction to Computing (OOP & Java)

Aug. 2024 – Dec. 2024

NC State University

- Mentored 100+ undergraduates in object-oriented programming, algorithm analysis, and debugging; held weekly office hours and graded assignments with structured, individualized feedback.

STARS AI Scholar: K–12 AI-Literacy Outreach

Dec. 2023 – Sep. 2025

STARS Computing Corps, NC State University

- Co-designed and field-tested an equitable AI/robotics curriculum for 50+ K–12 students with disabilities to evaluate scalable methods for broadening participation in computing.
- Developed and deployed accessible AI-robotics-instructional strategies across a 5th-grade pilot cohort, co-authored a peer-reviewed publication on scalable framework deployment [2].

VENTURE DEVELOPMENT & LEADERSHIP

Venture Research Lead

May 2026 – Jul. 2026

Associated Credit Management, Hong Kong SAR

- Executed instructional-design market analyses and user-retention profiling of regional edtech environments to provide strategic direction for new consumer ventures.
- Synced structural platform features (AI valuation metrics, escrow protocols) with regional regulatory reporting criteria.

Treasurer & Fundraising Chair

Aug. 2024 – Dec. 2025

Grand Challenges Student Organization, NC State University

- Restructured financial infrastructure to transition the cross-disciplinary engineering organization into an official, self-sustaining nonprofit entity.

HONORS & AWARDS

- STARS Award | STARS Computing Corps (2025)
- Robert E. Gross / Lockheed Aircraft Corporation Scholarship | NC State College of Engineering (2025)
- Connections Award | ASA DataFest (2025)
- Dean's List | NC State University (2023, 2024)
- Magna Cum Laude | University at Albany, SUNY (2022)

RESEARCH ETHICS & TRAINING

- CITI Program certifications in Human Subjects Research and the Responsible Conduct of Research (2024–2025)
- Developing Cultural Competence certification (2024)

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, HTML, CSS, C, Perl

AI/ML: LLM Evaluation and Verification, Prompt Engineering

Tools & Databases: Figma, Git, REST APIs, MySQL, Supabase, Docker, Nginx, USGS API, Postman, Apache, Expo Go

Frameworks: React, React Native, Next.js, Tailwind CSS, FastAPI, SQLAlchemy, Pandas, Spring, Hibernate, jQuery, Bootstrap